

## Worksheet: Material balances on a stripping column

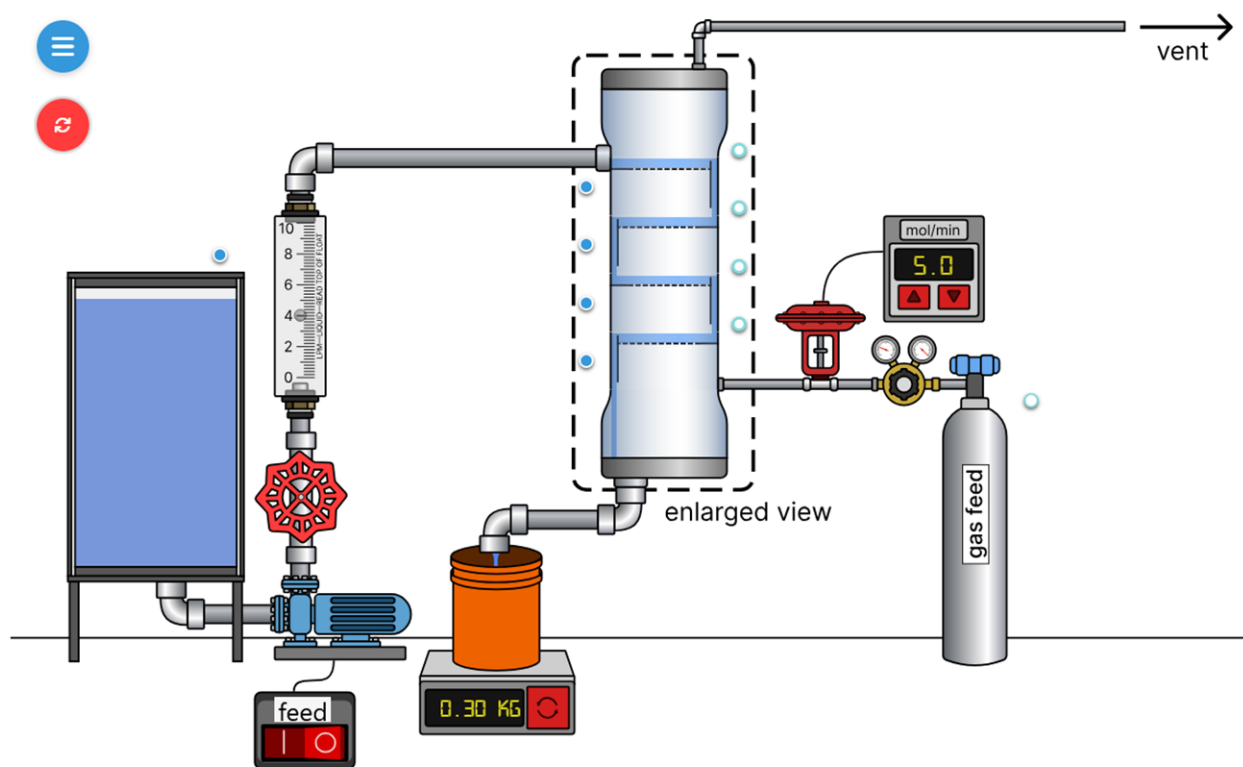
Name(s) \_\_\_\_\_

This experiment applies material balances to a trayed stripping column that removes a dilute solute from a downward-flowing liquid stream by stripping the solute into an upward-flowing gas stream. The column operates at steady-state at constant pressure.

### Student learning objectives

1. Be able to apply material balances to a system.
2. Be able to explain what happens in a stripping column.
3. Be able to apply the relation between gas-phase mole ratio and liquid-phase mole ratio (Henry's law).

### Equipment



The system consists of stages where the liquid phase (water) flows down, and the gas phase flows up. The phases leaving each stage are in phase equilibrium. The top stage is stage 1 and stage numbers increase moving down the column. Because the solute concentrations are low, Henry's law ( $y_i P = H x_i$ ) describes phase equilibrium, and the liquid and gas flow rates are approximately constant from stage to stage.

## Notation

$x_i$  = solute mole ratio in liquid leaving stage  $i$

$y_i$  = solute mole ratio in gas leaving stage  $i$

$L$  = liquid molar flow rate

$V$  = gas molar flow rate

$P$  = absolute pressure

$H$  = Henry's constant (bar)

Note that the solute mole ratios are in ppm.

## Questions to answer before starting the experiment

1. How many independent material balances can you write for the stripping column? Explain.
2. The two phases contact each other in a counter-current flow. What might be the advantage of counter-current flow over co-current flow?
3. Where is the highest concentration of solute in the stripping column, at the top or the bottom of the column? Why?

## Experimental Procedure:

1. Use the menu on the top left to select the stripping column mode and select 5 as the number of stages.
2. Switch on the liquid feed pump (left) to fill the column. Then, drag and rotate the red feed valve to change the liquid flow rate. Record the inlet volumetric flow rate, then calculate the mass flow rate, and enter the values in Table 1.
3. After liquid appears at the column outlet, the gas cylinder valve turns blue. Then, click on the blue valve to open it and pressurize the line. Rotate the pressure regulator valve to select a pressure. It may be easier to adjust the pressure regulator by zooming with the scroll wheel. Record the pressure at the top of Table 3. Use the buttons on the flow controller to adjust the gas flow rate and record the value in Table 2.
4. Allow the system to reach steady state. View liquid and vapor solute mole ratios by hovering over the blue dots (liquid) and white dots (gas) next to the column. Assume steady state if the solute mole ratios change by less than 2% in 30 s.
5. Using the weight scale at the liquid outlet, record the starting weight in Table 1 and start a timer. After some time has elapsed, record the weight and time in Table 1. Calculate the

liquid mass flow rate out and record it in Table 1. Calculate the difference between inlet and outlet liquid flow rates in Table 1.

6. Enter the liquid flow rate in g/min in Table 2 and calculate the flow rate in mol/min; the liquid is water.

Table 1 Liquid flow rate

| Volumetric flow rate in (L/min) | Mass flow rate in (kg/min) | Starting weight (kg) | Stopping weight (kg) | Elapsed time (s) | Mass flow rate out (kg/min) | Mass flow rate difference (in – out) kg/min |
|---------------------------------|----------------------------|----------------------|----------------------|------------------|-----------------------------|---------------------------------------------|
|                                 |                            |                      |                      |                  |                             |                                             |

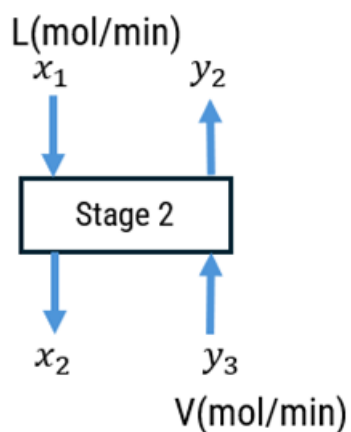
Table 2 Liquid and gas flow rates

| L(g/min) | L(mol/min) | V(mol/min) |
|----------|------------|------------|
|          |            |            |

7. Move the mouse over the buttons above the liquid and vapor feeds and record their solute mole ratios in Table 3. Likewise, record the solute mole ratios at the exit of each stage in Table 3. The figure for stage 2 shows what the subscripts refer to in the stage.

Table 3 Solute mole ratios      Pressure \_\_\_\_\_ bar

| Stage       | Solute mole ratio in liquid (ppm) | Solute mole ratio in gas (ppm) |
|-------------|-----------------------------------|--------------------------------|
| Liquid feed | $x_0 =$                           |                                |
| 1           | $x_1 =$                           | $y_1 =$                        |
| 2           | $x_2 =$                           | $y_2 =$                        |
| 3           | $x_3 =$                           | $y_3 =$                        |
| 4           | $x_4 =$                           | $y_4 =$                        |
| 5           | $x_5 =$                           | $y_5 =$                        |
| Gas feed    |                                   | $y_6 =$                        |



where  $L$  = liquid molar flow rate  
 $V$  = gas molar flow rate

### Data Analysis:

Carry out mole balances on the solute for each stage to determine if mole balances are satisfied. Table 4 shows the molar flow rates used for stage 2. Enter the mole balance values in Table 5.

Table 4 Notation for calculating mole balances

|       | Solute in ( $\mu\text{mol/min}$ ) |        |               | Solute out ( $\mu\text{mol/min}$ ) |        |               | Difference                      |              |
|-------|-----------------------------------|--------|---------------|------------------------------------|--------|---------------|---------------------------------|--------------|
| Stage | Liquid                            | Gas    | Total         | Liquid                             | Gas    | Total         | (In - out)                      | % difference |
| 2     | $x_1L$                            | $y_3V$ | $x_1L + y_3V$ | $x_2L$                             | $y_2V$ | $x_2L + y_2V$ | $(x_1L + y_3V) - (x_2L + y_2V)$ |              |

Table 5 Solute mole balance

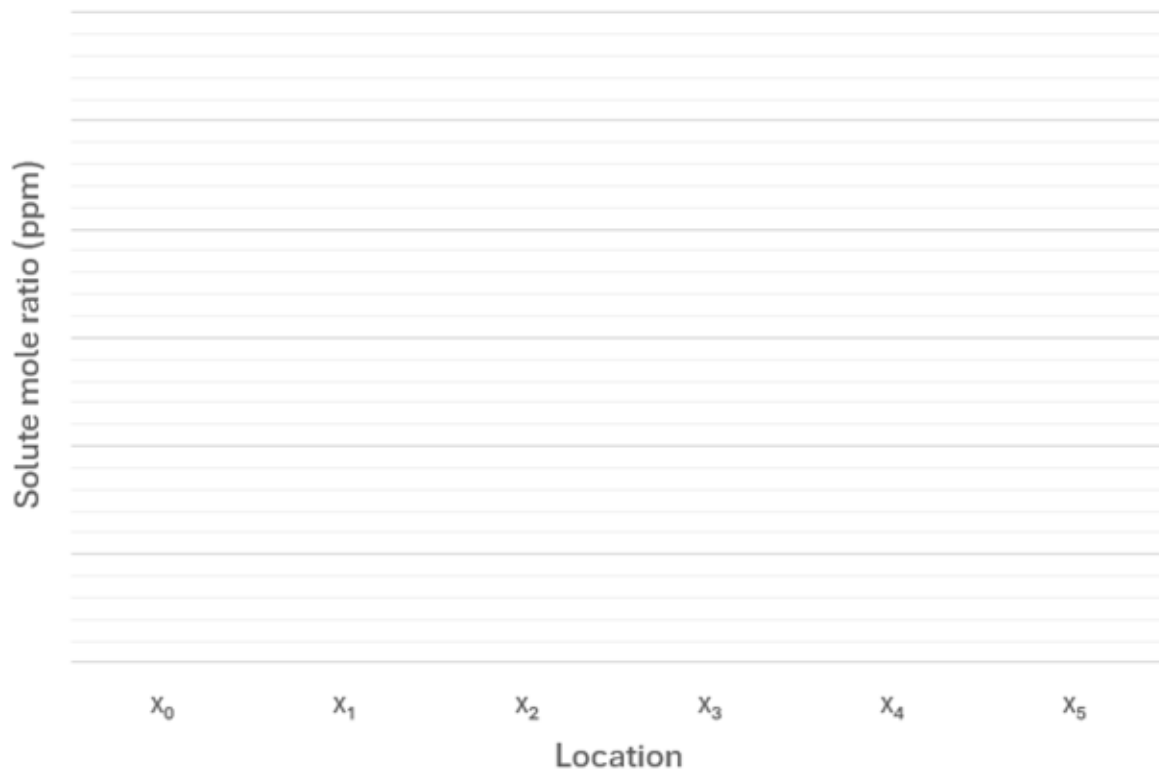
|       | Solute in ( $\mu\text{mol/min}$ ) |     |       | Solute out ( $\mu\text{mol/min}$ ) |     |       | Difference |              |
|-------|-----------------------------------|-----|-------|------------------------------------|-----|-------|------------|--------------|
| Stage | Liquid                            | Gas | Total | Liquid                             | Gas | Total | (In - out) | % difference |
| 1     |                                   |     |       |                                    |     |       |            |              |
| 2     |                                   |     |       |                                    |     |       |            |              |
| 3     |                                   |     |       |                                    |     |       |            |              |
| 4     |                                   |     |       |                                    |     |       |            |              |
| 5     |                                   |     |       |                                    |     |       |            |              |

The streams leaving each stage (e.g.,  $x_2$  and  $y_2$  for stage 2) are in equilibrium. Determine Henry's law constant for each stage from the equation  $y_i P = H x_i$  and enter the values in Table 3.

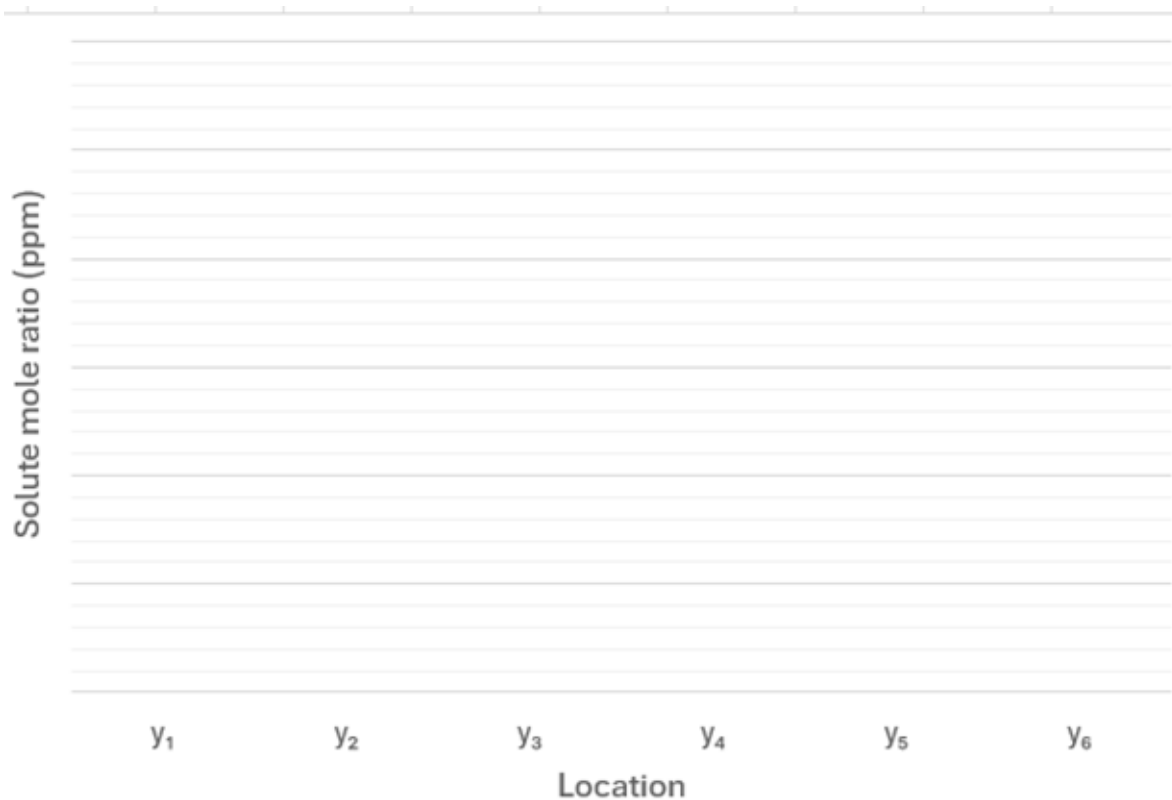
Table 6 Henry's law constant

| Stage   | H (bar) |
|---------|---------|
| 1       |         |
| 2       |         |
| 3       |         |
| 4       |         |
| 5       |         |
| average |         |

Plot the solute mole ratios for the liquid phase and connect the points with lines to clearly show the trends. The  $x_0$  notation is the liquid feed mole ratio into stage 1.



Plot the solute mole ratios for the gas phase and connect the points with lines to clearly show the trends. The  $y_{n+1}$  notation, where n is the number of stages, is the gas feed mole ratio into stage n.



**Questions to answer**

1. Where might these measurements have errors?
2. What safety precautions would you take to conduct this experiment in the laboratory?
3. To make the process more effective, would you run at low pressure or high pressure? Why? Change the pressure in the experiment and observe the effect on mole ratios.
4. To make this process more effective, would you run at low temperature or high temperature? Why?